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Data Governance for Sustainable AI in Organizations: A Benchmarkability-First Capability Model, Evidence Map, and Marketplace Microdata Demonstration

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Abstract

Sustainable AI requires more than ethical principles: organizations need measurable, auditable governance that links data controls, model lifecycle evidence, and operational monitoring to sustainability outcomes. This paper advances a measurement-first approach by (i) defining a 12-capability Data Governance for Sustainable AI (DG-SA) model with implementable evidence artifacts, (ii) mapping 18 influential standards, academic artifacts, and corporate frameworks to DG-SA using a conservative evidence-based coding protocol, and (iii) quantifying an open benchmarkability ratio over 54 governance survey items. Empirically, only 13 of 54 items (24.1%) admit defensible public/open-data proxy benchmarking, revealing a practical gap between what governance frameworks prescribe and what can be benchmarked externally. To address this gap, we incorporate a marketplace microdata pathway and demonstrate—using publicly available energy microdata distributed via cloud marketplaces—how joining governed AI telemetry with region-specific electricity carbon-intensity data yields audit-ready sustainability KPIs. In an illustrative cloud-region comparison for 2024, carbon intensity varies by approximately 17× across common regions, underscoring the need for governance-grade measurement conventions. We further provide a lifecycle operating model, dashboard blueprint, and minimum evidence pack (dashboards, runbooks, and decision gates) to make DG-SA actionable in production settings. The resulting artifacts translate sustainable AI governance from principle lists into testable, reproducible constructs and operational controls.

I. INTRODUCTION

Organizations are rapidly operationalizing AI for decision support, automation, and user-facing experiences. At the same time, regulators and stakeholders increasingly expect AI to be safe, accountable, and environmentally responsible. While ethical AI principles are widely published, they often do not translate into measurable practices that can be audited and improved over time. This gap is especially visible for sustainable (or Green) AI, where energy and carbon impacts are discussed extensively but measured inconsistently across organizations [1], [2].

Data governance provides a natural operational foundation for sustainable AI because AI systems are ultimately built on data pipelines, data controls, and evidence trails. Classical data governance literature emphasizes decision rights and control mechanisms [3], [4]. Contemporary AI governance standards extend these ideas with risk assessment, monitoring, and incident response expectations (e.g., NIST AI RMF [5], ISO/IEC guidance [6], [7]). Regulatory regimes such as GDPR [10] and the EU AI Act [11] further raise the bar for transparency, auditability, and lifecycle accountability.

However, two practical problems remain under-addressed. First, governance guidance is uneven: some capabilities are thoroughly specified (e.g., privacy and auditability), while others that matter for sustainable AI (e.g., energy measurement and drift surveillance) are treated lightly. Second, even when governance practices are recommended, they are difficult to benchmark externally because public and open data rarely provide comparable measures. Without benchmarkability, governance often becomes a compliance checklist rather than an improvement loop.

This work introduces a measurement-first governance approach for sustainable AI. It contributes not only a capability model, but also an evidence map and an explicit analysis of what can (and cannot) be benchmarked using public/open data. This combination is designed to move the topic from policy-level description to research-grade, testable constructs and empirical analysis.

Research questions (RQs) guiding the study are:

RQ1: What minimum set of governance capabilities is required to operationalize sustainable AI in organizations?

RQ2: How well do influential standards, academic artifacts, and corporate/platform frameworks cover these capabilities?

RQ3: To what extent can governance capabilities be benchmarked using authenticated public/open data proxies and primary marketplace datasets?

RQ4: What lifecycle operating model and minimum evidence pack make these capabilities actionable in engineering delivery and operations?

Contributions:

C1: DG-SA, a 12-dimension capability model with evidence-oriented definitions and implementable artifacts.

C2: An evidence map of 18 sources (6 standards/policy, 6 academic artifacts, 6 corporate/platform sources) coded against DG-SA.

C3: An open benchmarkability analysis of 54 survey items, showing that only 13/54 items (24.1%) admit external benchmarking via authenticated public proxies.

C4: A lifecycle operating model, dashboard blueprint, and minimum evidence pack that translate governance requirements into delivery and operations.

Novelty statement:

- **Benchmarkability-first governance**: DG-SA is explicitly defined in terms of observable evidence, and we operationalize governance progress via an open benchmarkability ratio rather than principle compliance.
- **Marketplace microdata augmentation**: we empirically demonstrate how cloud-marketplace microdata (electricity carbon intensity) can be joined with governed AI telemetry to produce audit-ready sustainability KPIs that are otherwise weakly benchmarkable via public surveys.
- **Operationalization**: we provide a lifecycle operating model and minimum evidence pack that converts sustainable AI governance into implementable engineering controls (gates, dashboards, and runbooks).

II. RELATED WORK AND CONCEPTUAL FOUNDATIONS

This section synthesizes four streams of work that motivate DG-SA: data governance as organizational control, AI governance as risk and assurance, sustainable AI measurement, and evidence-based benchmarking.

A. Data governance as control and accountability. IT and data governance research emphasize decision rights, accountabilities, and measurable controls as prerequisites for reliable enterprise systems [3], [4]. In AI settings, these ideas extend beyond data stewardship to include traceability of model decisions and accountability for downstream impacts.

B. AI governance frameworks and assurance artifacts. Major standards' bodies and policy institutions provide AI governance guidance, including NIST AI RMF [5], ISO/IEC risk management guidance [6], ISO/IEC AI management systems [7], and international policy principles [8], [9]. Industry frameworks, such as the Microsoft Responsible AI Standard [20] and IBM's Everyday Ethics for AI resources [22], complement standards with organization-level process controls and templates. Academic work argues that principles alone are insufficient without enforceable mechanisms and evidence [18]. Complementary artifact-based approaches such as Datasheets for Datasets [14] and Model Cards [15]-provide practical documentation structures, while algorithmic auditing frameworks formalize internal assurance processes [16], [17].

C. Sustainable AI and measurement. Green AI calls for efficiency and transparent reporting of AI resource use [1]. Empirical studies highlight energy and policy trade-offs in model development and inference [2], propose systematic reporting practices for energy and carbon footprints [12], [13]. Yet, sustainability metrics are often measured inconsistently across organizations, limiting comparability and cross-sector learning.

D. Benchmarking and evidence-based governance. Benchmarking is central to continuous improvement, but governance benchmarkability requires measurable constructs and comparable data. Many governance indicators are internal by nature (e.g., incident response maturity), which limits public evaluation. This motivates an explicit benchmarkability analysis: a governance model should disclose which aspects can be openly benchmarked and which require internal evidence packs.

III. DG-SA CAPABILITY MODEL

DG-SA is defined as a capability model that specifies governance dimensions using evidence-oriented criteria. A capability is considered operational only when it can be demonstrated through artifacts, logs, tests, policies, or measurable controls rather than aspirational principles. The 12 DG-SA capabilities are provided in Table I.

TABLE I. DG-SA Capability Codebook

Capability dimension	Evidence-oriented definition (paraphrased)
DataQuality	Defined data quality dimensions, validation, and stewardship for AI-relevant datasets.
Lineage	End-to-end lineage and versioned retention for datasets, features, prompts, and model artifacts; reproducibility of the pipeline.
AccessControl	Role-based access, least privilege, segregation of duties for data/model operations.
Privacy	PII handling, consent, minimization, retention, and privacy-preserving techniques.
ModelDocumentation	Model cards, intended use, limitations, versioning, and change history.
EvaluationBenchmarking	Benchmark suites, stress tests, red-team tests, and acceptance thresholds.
MonitoringDrift	Runtime monitoring for performance, drift, bias, and anomaly detection with alert thresholds.
HumanOversight	Human-in-the-loop control mechanisms, approvals, escalation paths, and override mechanisms.
RiskAssessment	Formal risk classification, hazard analysis, and mitigation control mechanisms across lifecycle.
SustainabilityEnergy	Energy/carbon measurement, efficiency targets, and sustainable compute policies.
Auditability	Immutable logs, traceability, and evidence packs to support audits and accountability.
IncidentResponse	Incident detection, triage, containment, post-mortems, and continuous improvement loops.

A. DG-SA as an extension of classical data governance. DG-SA retains foundational data governance controls (e.g., data quality, privacy, access control) while adding AI-specific governance needs such as model documentation, evaluation benchmarking, drift monitoring, human oversight, and sustainability/energy management. Table III summarizes the foundational vs. AI-specific extension view, while Table II provides an overview of the study artifacts and datasets used in this work.

TABLE II. Study Artifacts and Datasets

Artifact	Description	Scope/size
Coded corpus	Secondary sources (standards, academic, corporate) coded against 12 governance capabilities	n = 18
Capability codebook	Operational definitions for the 12 capabilities used for coding	n = 12
Public/open benchmark mapping	Proxy indicators mapped to selected governance survey items from publicly available surveys/reports	n = 31 datapoints
Survey instrument items	54 governance items used to assess what is measurable from public benchmarks	n = 54
Benchmark mapping dataset (public/open)	31 proxy datapoints mapped to 13 survey items; includes URLs, dates, and notes	31 rows; 13 items; 7 sources
Coded corpus matrix	Binary coding matrix used for coverage analysis (18×12)	18 sources × 12 capabilities

TABLE III. DG-SA Capabilities: Foundational vs AI-Specific Extensions

Category	Capability	AI-specific rationale
Foundational DG	DataQuality	Quality controls, validation, and error handling for data feeding AI.
Foundational DG	AccessControl	Least privilege, RBAC, and data/model access enforcement.
Foundational DG	Privacy	PII minimization, lawful basis, and privacy-by-design controls.
Foundational DG	Auditability	Immutable logs, lineage, traceability, and audit trails.
Cross-cutting	RiskAssessment	Risk classification for AI use (incl. safety, bias, compliance).
AI-specific extension	ModelDocumentation	Model cards, intended use, limitations, training data provenance.
AI-specific extension	EvaluationBenchmarking	Pre-deploy and regression benchmarking with acceptance thresholds.
AI-specific extension	MonitoringDrift	Runtime surveillance of performance/quality drift and data shifts.
AI-specific extension	HumanOversight	Human-in-the-loop triggers, approvals, and override controls.
AI-specific extension	IncidentResponse	AI incident handling, escalation, containment, and postmortems.
AI-specific extension	Lineage	Versioned datasets/models, retention policies, reproducibility.
AI-specific extension	SustainabilityEnergy	Energy/carbon measurement, efficiency targets, and reporting.

B. Positioning relative to NIST/ISO governance. DG-SA is designed to be complementary to risk management and management systems standards. NIST AI RMF emphasizes high-level functions (govern–map–measure–manage) [5], while ISO/IEC 42001 formalizes management system requirements [7]. DG-SA focuses on capability-level evidence that can be implemented and measured, thereby providing an operational layer that can be mapped to these frameworks. Table IV provides a high-level mapping to clarify this relationship.

TABLE IV. Illustrative Mapping: DG-SA to NIST AI RMF and ISO/IEC 42001 Concepts

DG-SA capability	NIST AI RMF function(s)	ISO/IEC 42001 concept	Example evidence artifacts
RiskAssessment	MAP / MEASURE	Risk identification and measurement	AI impact/risk assessments, risk registers
Privacy	GOVERN / MANAGE	Policy + controls	PII controls, DPIA where applicable
Auditability	GOVERN / MANAGE	Accountability and traceability	Audit logs, trace IDs, evidence retention
ModelDocumentation	MAP	System context and documentation	Model cards, intended use + limitations
EvaluationBenchmarking	MEASURE	Testing and validation	Benchmark suites, acceptance thresholds
MonitoringDrift	MANAGE	Ongoing monitoring	Drift dashboards, alerts, rollback criteria
HumanOversight	GOVERN / MANAGE	Human accountability	HIL triggers, approvals, overrides
SustainabilityEnergy	MEASURE	Sustainability measurement	Energy/carbon reporting, efficiency KPIs

IV. RESEARCH DESIGN AND METHODS

The study is a mixed conceptual–empirical design. Conceptually, DG-SA is developed by consolidating governance and sustainable AI requirements into evidence-based capability definitions. Empirically, we (i) code a curated corpus of influential sources against DG-SA and (ii) test the public benchmarkability of governance survey items via open proxy mapping.

A. Corpus selection (coded evidence corpus). We constructed a corpus of 18 sources spanning standards/policy, academic research artifacts, and corporate frameworks. The selection objective was not exhaustive, but influence and coverage: documents were included if they are widely referenced, officially adopted, or provide reusable governance mechanisms for organizational AI. Searches and shortlisting were conducted across standards' repositories (NIST, ISO, OECD, UNESCO, EU), academic indexes (Google Scholar, IEEE Xplore, ACM DL), and corporate governance publications. Inclusion criteria required: (i) organizational applicability (controls, processes, or assurance artifacts), (ii) explicit relevance to AI governance or data governance in AI contexts, (iii) public accessibility, and (iv) sufficient specificity to evaluate evidence requirements.

To reduce selection bias, the coded corpus was balanced across source types (6 standards/policy, 6 academic artifacts, and 6 corporate/platform sources), selected for prominence and practical relevance to implementable governance. The corpus includes, for example, NIST AI RMF [5], ISO standards [6], [7], and international policy guidance [8], [9], [11]; academic evidence artifacts such as Datasheets and Model Cards [14], [15] together with auditing and verifiable-claims mechanisms [16], [17]; and corporate/platform operational guidance including the Microsoft Responsible AI Standard [20], Google's AI responsibility lifecycle [32], AWS responsible AI guidance [33], and cloud data-exchange documentation (AWS Data Exchange [34], Snowflake Marketplace [35], and BigQuery sharing/Analytics Hub [36]).

TABLE V. Coded Evidence Corpus (N = 18) and DG-SA Coverage Ratios

ID	Year	Type	Source title (short)	Coverage count (0–12)	Coverage ratio (%)
A1	2010	Academic	Designing data governance (Khatri & Brown) [4]	3/12	25.0
A2	2019	Academic	Model Cards for Model Reporting (Mitchell et al.) [15]	3/12	25.0
A3	2021	Academic	Datasheets for Datasets (Gebru et al.) [14]	3/12	25.0
A4	2020	Academic	Closing the AI accountability gap: internal algorithmic auditing (Raji et al.) [16]	10/12	83.3
A5	2020	Academic	Toward trustworthy AI development: mechanisms for supporting verifiable claims (Brundage et al.) [17]	10/12	83.3
A6	2020	Academic	Green AI (Schwartz et al.) [1]	3/12	25.0
C1	2022	Corporate	Microsoft Responsible AI Standard v2 (General Requirements) [20]	11/12	91.7
C2	2024	Corporate	End-to-end responsibility: A lifecycle approach to AI (Google) [32]	8/12	66.7
C3	2024	Corporate	AWS Responsible Use of AI Guide [33]	10/12	83.3
C4	2025	Corporate	AWS Data Exchange User Guide (Overview/Subscribing) [34]	2/12	16.7
C5	2026	Corporate	Snowflake Marketplace and Listings Documentation [35]	3/12	25.0
C6	2026	Corporate	BigQuery sharing (formerly Analytics Hub) Documentation [36]	3/12	25.0
S1	2023	Standards	NIST AI Risk Management Framework 1.0 [5]	11/12	91.7
S2	2023	Standards	ISO/IEC 23894:2023 AI risk management guidance [6]	10/12	83.3
S3	2023	Standards	ISO/IEC 42001:2023 AI management system [7]	9/12	75.0
S4	2019	Standards	OECD Recommendation on Artificial Intelligence [8]	2/12	16.7
S5	2021	Standards	UNESCO Recommendation on the Ethics of Artificial Intelligence [9]	5/12	41.7
S6	2024	Standards	EU Artificial Intelligence Act [11]	11/12	91.7

B. Coding protocol and operational decision rule. Each source was coded across the 12 DG-SA capabilities using a conservative, evidence-based rule: a capability was marked present (1) only when the source specified implementable controls or required

evidence (e.g., artifacts, tests, logs, documentation structures, monitoring expectations). Merely stating a principle without operationalization was coded as absent (0). This rule mitigates inflation of coverage that can occur when principle lists are treated as equivalent to implementation guidance. Coding was conducted at the document level (unit of analysis = source), with a maintained coding log to make decisions reviewable.

Inter-rater reliability and reproducibility: the present evidence map was produced by a single coder. To make the analysis reviewable, we provide (i) an operationalized codebook (Table I), (ii) the complete 18×12 binary coding matrix and source metadata (Appendix D), and (iii) explicit decision rules that prioritize implementable evidence over principle statements. These artifacts enable replication and independent multi-coder reliability assessment as a straightforward extension for future work (Section X).

C. Benchmarkability mapping protocol. We evaluated benchmarkability for 54 governance survey items (developed to measure DG-SA capabilities). For each item, we attempted to identify a public/open-data proxy that could serve as an external benchmark. Searches were performed between Dec. 2024 and Nov. 2025 using query patterns that combined each construct with terms such as "benchmark", "survey", "adoption rate", "coverage", "maturity", and "observability". A proxy was accepted when it (i) measured an organization-level practice/outcome close to the survey construct, (ii) provided an explicit numeric or clearly bounded estimate, (iii) identified the source organization and reporting year, and (iv) allowed documentation of scope and assumptions. Each mapping was assessed as direct or approximate (Appendix A2), reflecting conceptual distance between the proxy and the survey item.

Publication bias is a known risk in open benchmarking: public surveys tend to over-represent large organizations and consultancies. We therefore treat the benchmarkability dataset as exploratory and emphasize transparent reporting of proxy limitations. The benchmark dataset uses sources such as ISACA [23], Precisely [24], EY [25], Dresner [26], TDAN [27], Fastly [28], the AI Incident Database [29], and McKinsey [30], [31].

Proxy provenance/authentication. Public proxy statistics are frequently re-posted across reports and blogs, which can obscure the original survey instrument, sample, and fielding period. To strengthen validity, we authenticate each public proxy against its primary source wherever possible (e.g., the original report PDF or survey owner's publication) and record (i) sampling frame and respondent profile, (ii) sample size, geography, and fielding dates, (iii) question wording (when available), and (iv) licensing/reuse constraints. We summarize this provenance information and assign an authentication tier in Appendix C (Table C2).

Primary-data pathway via data exchanges. Where public proxies are absent or too coarse, we propose augmenting benchmarkability using primary datasets accessible through cloud data exchanges and marketplaces-including AWS Data Exchange [34], Snowflake Marketplace [35], and BigQuery sharing/Analytics Hub [36]-as well as comparable marketplaces (e.g., Databricks Marketplace [37] and Azure Data Share [38]). These exchanges enable licensed access to structured organizational and ecosystem data (e.g., ESG disclosures, corporate filings, job postings, vulnerability feeds, and energy/carbon intensity datasets) that can operationalize DG-SA indicators with higher resolution and auditability (Appendix C, Table C1). Section VIII provides an empirical demonstration of this pathway using marketplace-distributed electricity carbon-intensity microdata (OWID/Ember) joined with governed AI telemetry to compute an auditable footprint KPI.

TABLE VI. DG-SA Dashboard Blueprint (Illustrative)

Dashboard layer	Example indicators	Decision trigger (example)
Governance compliance	Artifact completeness (%), model card coverage, dataset lineage present/absent	Block release if required evidence pack is incomplete
Runtime trust & safety	Drift score, policy violations, incident rate, human override rate	Escalate to review board if incident rate exceeds threshold
Sustainability & efficiency	Energy per training run, cost per 1K inferences, utilization, carbon intensity (if available)	Require efficiency review if energy per inference increases > X% QoQ

TABLE VII. Benchmarkable Survey Items and Proxy Coverage (N = 13)

Question ID	Survey item (short)	# Sources	# Datapoints
Q10	Formal data governance exists in my organization (single select): Yes; In progress; …	3	3
Q11	Formal AI governance exists in my organization (single select): Yes; In progress; No.	2	3
Q13	We have an ethics review board / model risk committee that reviews AI use cases (Yes…	1	2
Q16	Estimated monitoring coverage across deployed AI systems (0–100%).	1	2
Q19	Cross-functional training frequency for governance/responsible AI topics (single sel…	1	3
Q25	Our organization has a defined ethical AI framework that is used in delivery.	1	1
Q39	Energy use is measured per training/inference job.	1	1
Q4	AI adoption stage (single select): Exploring; Pilot; Production; Scaling.	1	3
Q41	Green training practices are used (e.g., efficient architectures, scheduling, hardwa…	1	1
Q46	Estimated reduction in data quality issues after governance (%).	1	1
Q48	Estimated reduction in training energy after efficiency/governance interventions (%).	1	2
Q51	Alignment of AI work with sustainability targets (Likert 1–5).	1	3
Q52	Top challenge (single select): Fragmented standards; Skill gaps; Tooling gaps; Data …	3	6

D. Statistical treatment. Our analysis is primarily descriptive. Coverage counts and ratios are computed per capability and per source. To explore whether coverage improves over time, we compute Pearson correlation between publication year and source coverage ratio, reporting r, p-value, and a 95% confidence interval; this is explicitly treated as exploratory because N = 18 is small. For benchmarkable items, we report proxy ranges and medians (Fig. 6) and provide numeric summaries in text to avoid purely qualitative range descriptions.

V. RESULTS

A. Capability coverage across the coded corpus. Figure 3 summarizes how frequently each DG-SA capability is operationalized across the 18-source corpus. Auditability is the most consistently evidenced capability (15/18 sources), followed by Privacy (13/18) and Human Oversight (12/18). Data Quality and Risk Assessment are each evidenced in 11/18 sources, while Access Control, Model Documentation, and Evaluation/Benchmarking each appear in 10/18 sources. Monitoring/Drift and Incident Response are each operationalized in 9/18 sources. Two capability dimensions remain notably under-specified: Lineage (5/18) and Sustainability/Energy (2/18). This pattern suggests that most influential frameworks provide substantial guidance for compliance-critical controls, but far fewer provide implementable evidence for end-to-end traceability and environmental sustainability.

B. Coverage by source type. Figure 4 reports the mean DG-SA coverage ratio by source type. Standards/policy sources have the highest mean coverage (66.7%), followed by corporate/platform sources (51.4%), and academic artifacts (44.4%). The higher standards coverage is driven by comprehensive, multi-domain requirements (e.g., NIST AI RMF and the EU AI Act), while principle-only sources contribute fewer evidence-oriented controls. Academic artifacts tend to be deep in a small subset of capabilities (e.g., documentation, auditing, or sustainability), rather than broad across the full DG-SA set.

C. Temporal trend (exploratory). Figure 5 plots DG-SA coverage ratio against publication year. The Pearson correlation is $r = 0.22$ ($n = 18$, $p = 0.372$; 95% CI [-0.27, 0.63]). The direction is weakly positive but not statistically distinguishable from zero at conventional thresholds, reinforcing that gaps (especially sustainability/energy and lineage) persist even in recent governance sources.

D. Open benchmarkability ratio. Table VII reports that 13 of 54 governance survey items (24.1%) can be mapped to public/open-data proxies with sufficient specificity for external benchmarking. To avoid treating secondary re-publications as primary evidence, we authenticate proxy sources where possible and record sampling and methodological metadata (Appendix C). The low benchmarkability ratio indicates that most governance constructs are currently measured internally (or qualitatively), rather than through externally observable indicators; this motivates the data-exchange primary-data pathway described in Section IV.C and Appendix C.

E. Range behavior for benchmarkable items. For items with percent-like numeric proxies (Fig. 6), dispersion is substantial for several constructs (e.g., Q10 ranges from 31% to 71% across sources). Even when medians are stable, ranges often reflect differences in survey framing, geography, and respondent roles. This reinforces the need to report provenance and to prefer primary microdata (or traceable marketplace datasets) when converting survey results into benchmark indicators.

TABLE VIII. Percent-like Proxy Summary for Selected Benchmarkable Survey Items

Question ID	# proxies	Min	Median	Max
Q10	3	31.0	32.0	71.0
Q11	2	28.0	29.5	31.0
Q13	2	61.0	67.5	74.0
Q16	2	9.5	10.2	11.0
Q19	3	22.0	32.0	35.0
Q39	1	67.0	67.0	67.0
Q46	1	68.0	68.0	68.0
Q48	2	30.0	40.0	50.0
Q51	3	28.0	36.0	45.0

F. Sustainability proxy scarcity. SustainabilityEnergy is the least represented DG-SA capability in the coded corpus (11.1%). Consistently, the benchmarkability dataset contains only limited public proxies for energy/footprint governance. For example, a Fastly report indicates that more than two-thirds of surveyed organizations track training and inference energy separately (Q39), but such statements rarely include standardized measurement baselines or comparable denominators. This reinforces the conclusion that sustainable AI governance needs stronger measurement conventions, and that open benchmarking in this area remains early-stage [1], [12].

VI. SYNTHESIS AND DISCUSSION

The evidence map and benchmarkability analysis jointly suggest a practical interpretation: governance is increasingly well-specified for compliance-critical capabilities (privacy, auditability, risk assessment), but weaker for operational and sustainability-centric capabilities. The benchmarkability ratio adds an additional constraint-public/open benchmarking is possible for only about one-quarter of governance survey items, and even these require proxies that may be conceptually approximate.

Implication 1 (measurement-first governance)

organizations should treat public benchmarks as a complement, not a substitute, for internal evidence packs. External proxies can offer directional signals (e.g., typical governance adoption or monitoring coverage), while acceptance decisions for high-risk AI must rely on auditable internal evidence (tests, logs, red-team reports, model cards).

Implication 2 (AI-specific differentiation)

several DG-SA capabilities are explicitly AI-specific (ModelDocumentation, EvaluationBenchmarking, MonitoringDrift, HumanOversight, SustainabilityEnergy). These are precisely the areas where generic data governance is insufficient because

models evolve, drift, and interact with humans and environments. The coded corpus shows these capabilities are less consistently operationalized than privacy and auditability, suggesting a priority area for future standards work.

Implication 3 (sustainable AI as operational practice)

sustainable AI governance must be integrated into lifecycle operations, not treated as an after-the-fact reporting exercise. Evidence needs to include measurement methods, thresholds, and response playbooks (e.g., what action is taken when energy per inference rises above a budget).

VII. MINI-CASE INSTANTIATION: LLM-BASED BUSINESS ASSISTANT

To address application-oriented expectations, we instantiate DG-SA for a realistic enterprise automation case: a retrieval-augmented LLM assistant that answers customer queries and drafts compliant responses. The system consumes enterprise knowledge bases and ticket histories, performs retrieval and prompt orchestration (Data AIService), executes an LLM endpoint (Model AIService), applies policy/guardrail logic (Algo AIService), and exposes the capability via an API and web interface.

In this mini-case, DG-SA capabilities translate into concrete artifacts. For example, ModelDocumentation is realized via a model card documenting intended use, known failure modes, and training data constraints [15]. EvaluationBenchmarking requires a benchmark suite (including red-team prompts) and acceptance thresholds before release [16], [17]. MonitoringDrift is implemented as runtime dashboards tracking response quality, safety violations, and retrieval health. SustainabilityEnergy becomes actionable when the assistant reports energy per request and enforces compute budgets, aligned to Green AI reporting practices [1], [12].

TABLE IX. Mini-Case: Example DG-SA Evidence Artifacts and Lifecycle Use

DG-SA capability	Primary artifact	What it evidences	Where used
ModelDocumentation	Model card + change log	Model choice, version, limitations	Release gate (DoD)
EvaluationBenchmarking	Benchmark + red-team report	Accuracy, safety, latency thresholds	Release gate (DoD)
MonitoringDrift	Quality & drift dashboards	Retrieval hit-rate, hallucination indicators	Operate + rollback
IncidentResponse	AI incident runbook	Safety incident triage, escalation	Operate
SustainabilityEnergy	Energy/footprint report	Energy per request, budget adherence	Operate + scale decision

VIII. MARKETPLACE MICRODATA DEMONSTRATION

Public/open proxies are often too coarse to benchmark sustainability-oriented AI governance. This section provides a concrete empirical demonstration of how cloud data marketplaces can supply primary microdata that, when joined with governed AI telemetry and lineage, yields audit-ready sustainability indicators.

A. Dataset and provenance. We use the Our World in Data (OWID) energy dataset, which includes a country-year series for the lifecycle carbon intensity of electricity (carbon_intensity_elec, in gCO_2e/kWh). OWID curates these series from primary energy-system sources (e.g., Ember) and distributes the dataset both directly and through data-marketplace channels (e.g., Snowflake data sharing). For the illustration below, we use 2024 values to reflect recent grids while avoiding partial-year artifacts. [39], [40].

B. Demonstration 1: Carbon-intensity dispersion across illustrative cloud regions. To highlight why SustainableAI/Energy governance must capture region metadata, we map a set of commonly used cloud regions to their host countries and compare 2024 electricity carbon intensity. Fig. 8 shows a wide dispersion: across these regions, carbon intensity ranges from approximately 41 gCO_2e/kWh (France) to approximately 707 gCO_2e/kWh (India), a ≈17× spread. This implies that identical AI workloads can have order-of-magnitude different operational emissions solely due to regional electricity mix, making location-aware telemetry and standardized reference data a governance requirement—not an optional sustainability add-on. [39].

C. Demonstration 2: Marketplace microdata join for an auditable AI footprint KPI. DG-SA treats sustainability as measurable control. A minimal auditable KPI is per-model or per-release emissions, computed as: Emissions ($kgCO_2e$) = $\Sigma_{r,t} E_{r,t}$ (kWh) × $CI_{r,t}$ (gCO_2e/kWh) / 1000, where $E_{r,t}$ is governed AI energy telemetry (or GPU-hours converted to kWh) and $CI_{r,t}$ is carbon-intensity microdata for the execution region and time. Fig. 9 summarizes the join pattern: an external carbon-intensity feed becomes decision-grade once connected to internal telemetry, model documentation, and lineage under governance control.

D. Implications for benchmarkability. The demonstration operationalizes the SustainabilityEnergy capability as an externally auditable indicator, even when public benchmarking surveys provide only high-level adoption percentages. In practice, organizations can standardize reporting (e.g., per-model emissions per 1,000 inferences; per-training-run emissions with uncertainty bounds) and publish comparable statistics using a combination of internal evidence and marketplace-sourced reference microdata. This provides a pragmatic route to closing the measurement gap identified in Section V.

IX. LIFECYCLE OPERATING MODEL AND MINIMUM EVIDENCE PACK

DG-SA is intended for forward-engineering delivery. We therefore propose a lifecycle operating model that separates four planes-data, model, runtime, and feedback-so that evidence can be produced and accepted at gates. This helps reconcile two realities: (i) governance must be enforced during build, integration, and operations; and (ii) sustainable AI requires continuous measurement and improvement loops.

We define a minimum evidence pack to make acceptance decisions auditable. The evidence pack aligns to typical release gates (build, verification, UAT, go-live) and includes: model evaluation evidence, security and privacy controls, monitoring readiness, incident response runbooks, and sustainability reporting. Table X summarizes this minimum pack.

TABLE X. Minimum Evidence Pack for DG-SA Acceptance

Lifecycle phase	Minimum evidence artifacts (examples)	Gate decision (example)
Data onboarding	Datasheet, lineage record, access/privacy classification, quality checks	Reject dataset if lineage or privacy constraints are missing
Model development	Model card draft, evaluation plan, benchmark suite, risk register entry	Block training if evaluation plan is undefined
Pre-release validation	Benchmark results, robustness tests, red-team notes, monitoring plan, rollback plan	Release only if thresholds met and rollback defined
Production operation	Drift dashboard, incident logs, periodic audit report, sustainability metrics	Escalate if incidents/drift exceed thresholds

Research propositions (forward-looking): P1-Benchmarkable metrics will expand as observability and AI management systems mature, but sustainability metrics will lag unless standardized measurement conventions are adopted. P2-Evidence packs will become the dominant governance mechanism for high-risk AI because they bind requirements to auditable artifacts. P3-Organizations will increasingly integrate AI incident response with security incident response, supported by centralized audit logging and traceability. P4-AI sustainability will shift from voluntary reporting to budgeted operational constraints (e.g., energy per inference budgets) as compute costs and regulatory attention increase.

X. LIMITATIONS AND FUTURE WORK

Internal validity

the evidence map relies on a curated corpus (N = 18) and binary coding. Binary coding improves transparency but can under-represent partial coverage; this is a deliberate conservative choice.

External validity

the corpus is not exhaustive, and corporate/platform sources may reflect sector-specific operating models.

Reliability: the present version reports single-coder results, but provides the full coding matrix and decision rules (Appendix D) so that other researchers can replicate the coding and compute inter-rater reliability in follow-on studies. Benchmarkability limitations: public proxies are sparse, noisy, and biased toward organizations that publish surveys and reports; we mitigate this with a provenance/authentication rubric (Appendix C) and propose a primary-data pathway via data exchanges (Section IV.C).

Benchmarkability limitations

open proxies are sparse, noisy, and biased toward large organizations that publish surveys and reports. Proxy mappings may be approximate when constructs do not align perfectly. Future work should develop standardized open benchmarks for AI governance and sustainability, including common denominators for energy reporting (e.g., energy per inference normalized by workload).

Future research directions include

expanding the corpus across additional sectors and regions; developing ordinal or weighted coverage scoring to complement binary coding; instantiating DG-SA in multiple AI applications (e.g., credit scoring, medical triage) to test usability; and studying how DG-SA evidence packs influence operational outcomes such as incident rates, audit findings, and compute efficiency over time.

XI. CONCLUSION

This paper presented DG-SA, a measurement-first governance model for sustainable AI. By combining a 12-dimension capability model with an evidence map and an explicit open benchmarkability analysis, the study moves beyond descriptive governance narratives toward testable, auditable constructs. Empirically, we find that compliance-critical capabilities (privacy, auditability, risk assessment) are most consistently operationalized across influential sources, while sustainability/energy governance is least operationalized. We also show that only 24.1% of governance survey items are benchmarkable via public/open data proxies. Finally, we propose a lifecycle operating model and minimum evidence pack to support engineering delivery and operations. Collectively, these contributions support sustainable AI as an implementable organizational practice and provide a baseline for future empirical research in AI governance measurement.

Declarations

Funding: The authors received no specific funding for this work.

Competing interests: The authors declare no competing interests.

Data availability: All external datasets referenced in the empirical demonstration are publicly accessible from their primary publishers; derived analysis outputs (country-year carbon intensity subset, figures, and the mapping used in Fig. 8) are provided as supplementary material. The coded evidence matrix and proxy mapping tables are included in the appendices.

Code availability: The scripts used to generate the empirical figures and descriptive statistics are provided as supplementary material to support reproducibility.

Ethics approval and consent to participate: Not applicable; this study does not involve human participants or personal data.

Consent for publication: Not applicable.

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Figures

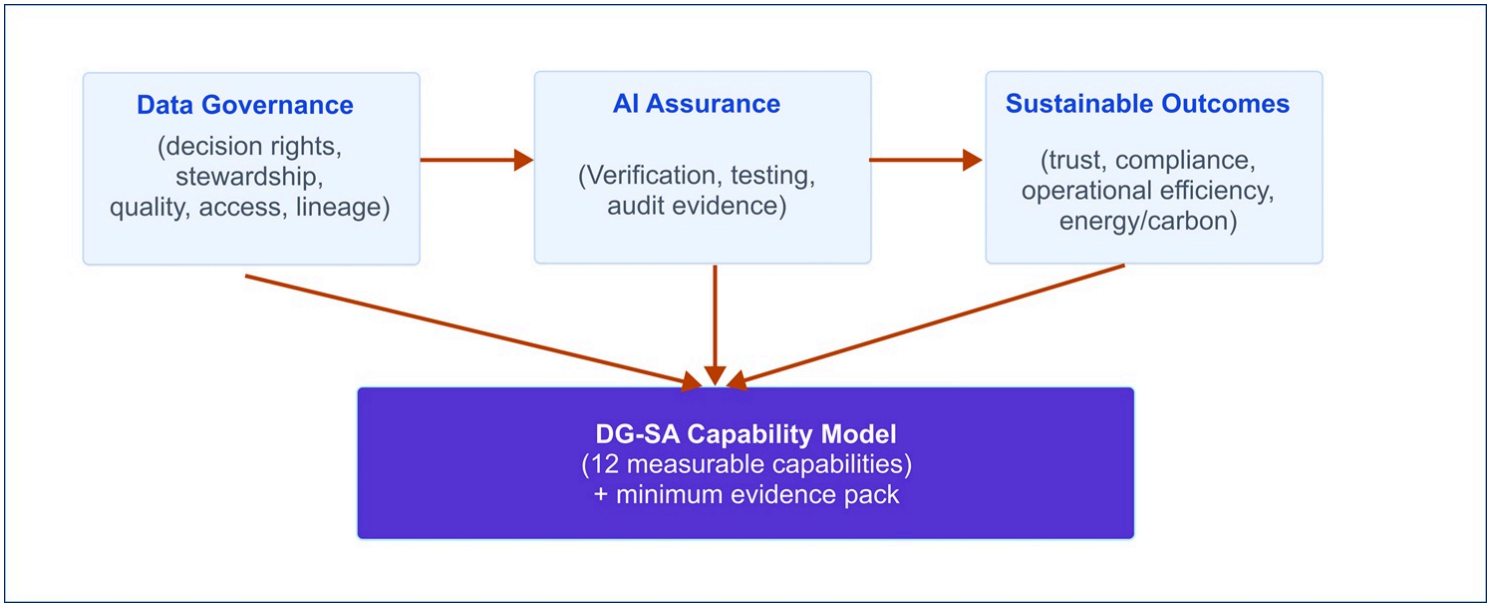

Figure 1

Conceptual framing: DG-SA connects data governance, AI assurance, and sustainable AI measurement.

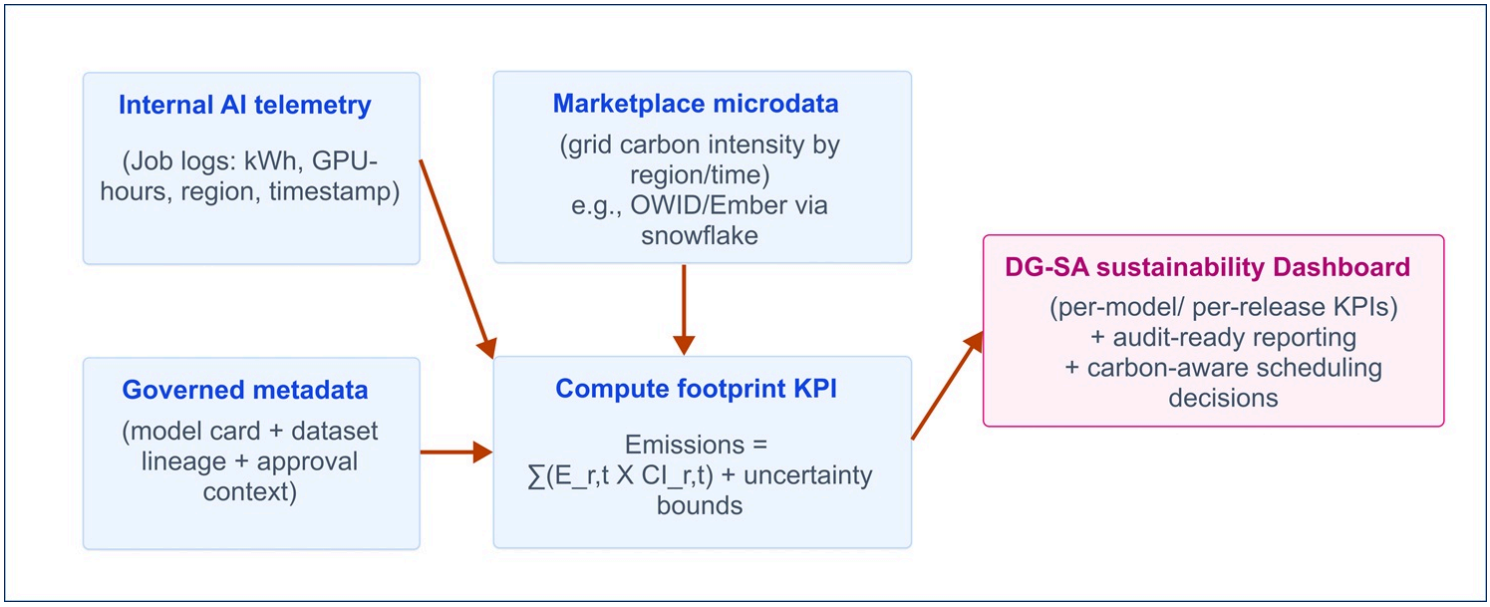

Figure 2

Research design workflow (conceptual development → coding → benchmarkability mapping → synthesis).

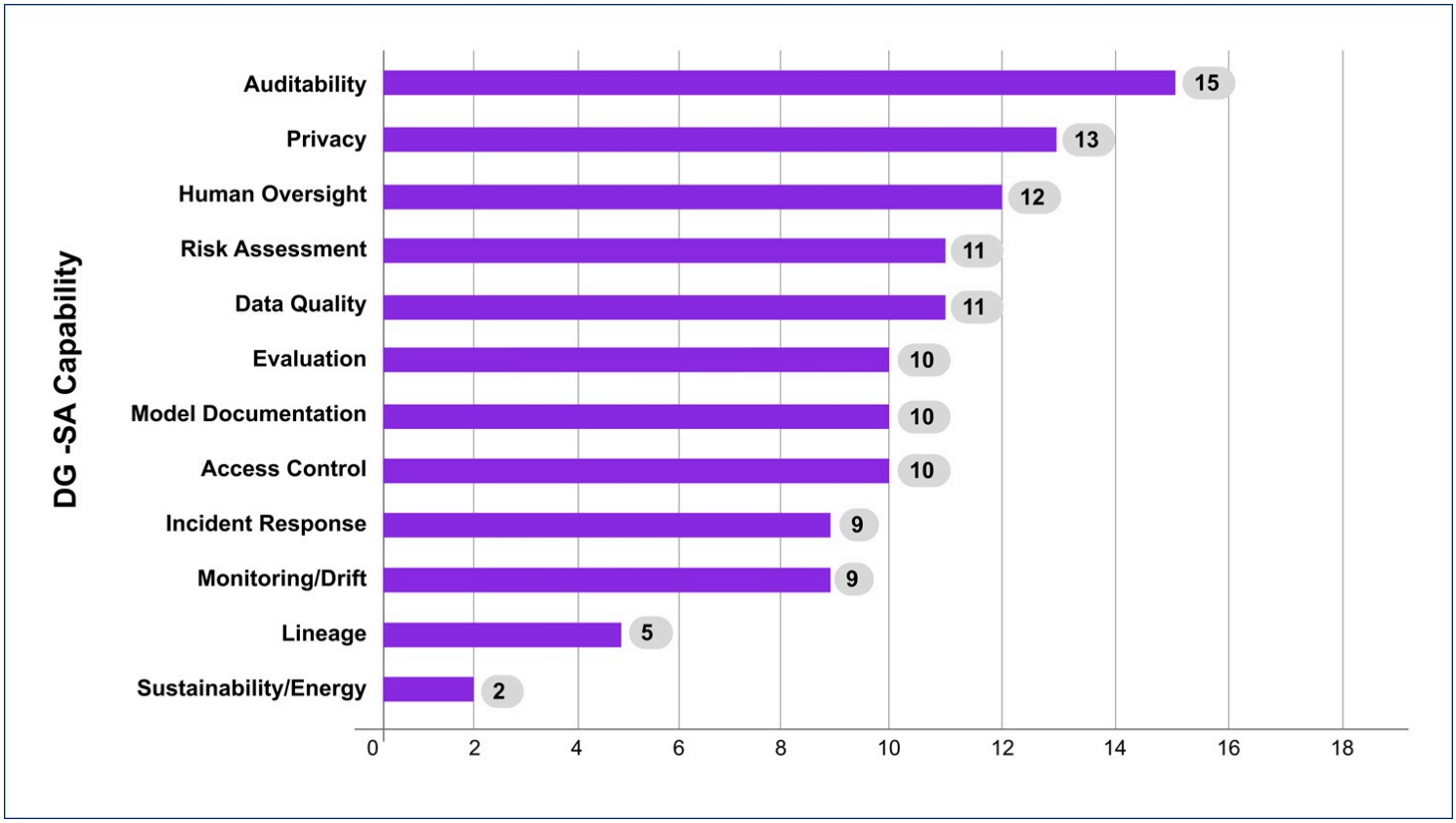

Figure 3

DG-SA capability coverage across 18 coded sources.

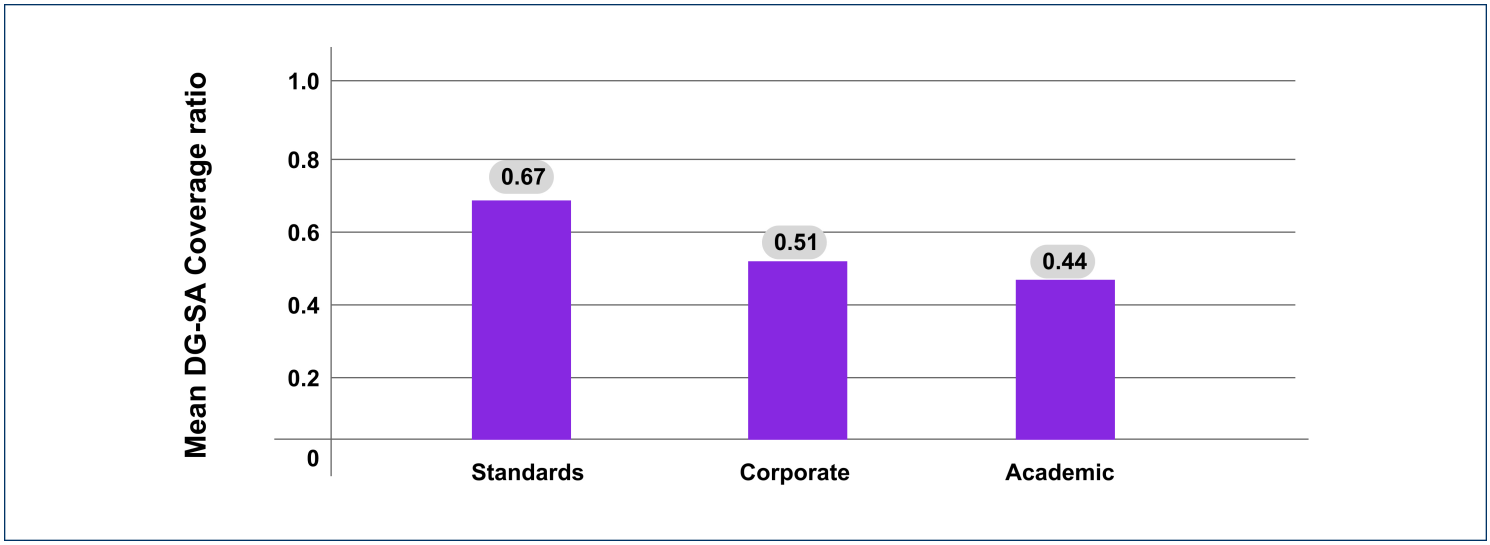

Figure 4

Mean DG-SA coverage ratio by source type.

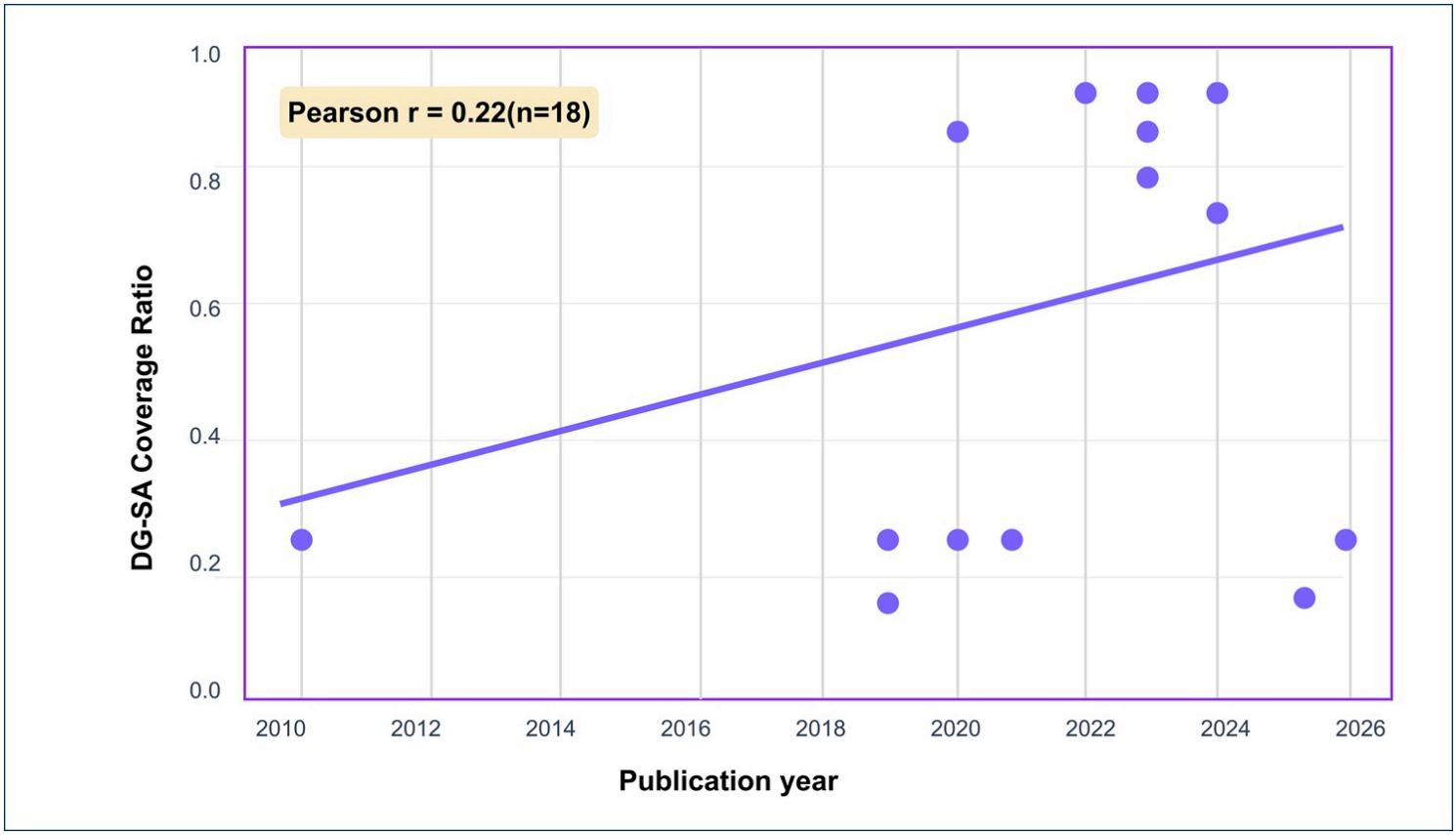

Figure 5

Publication year vs. DG-SA coverage ratio (exploratory correlation).

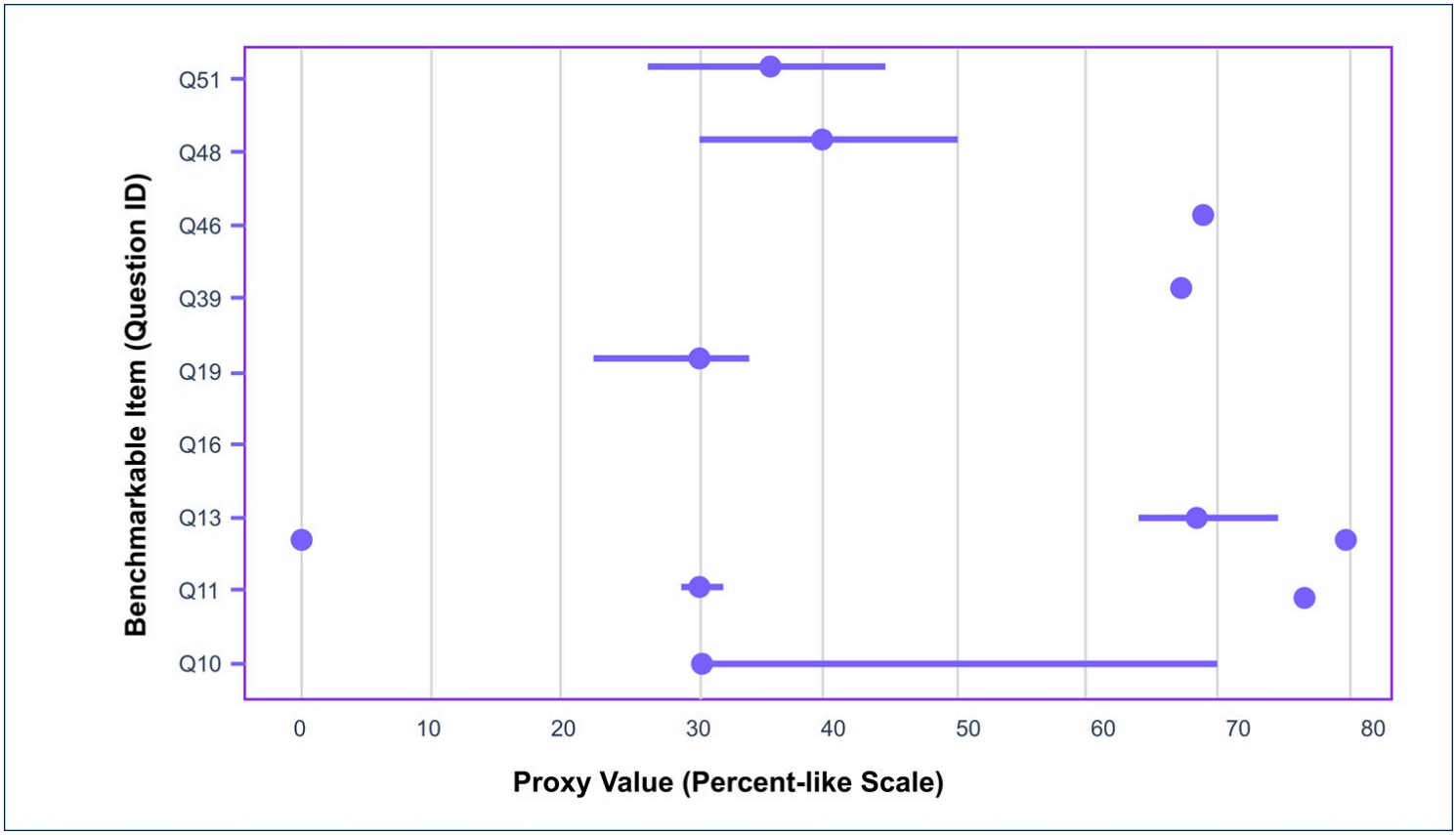

Figure 6

Ranges of percent-like open proxies for selected benchmarkable survey items.

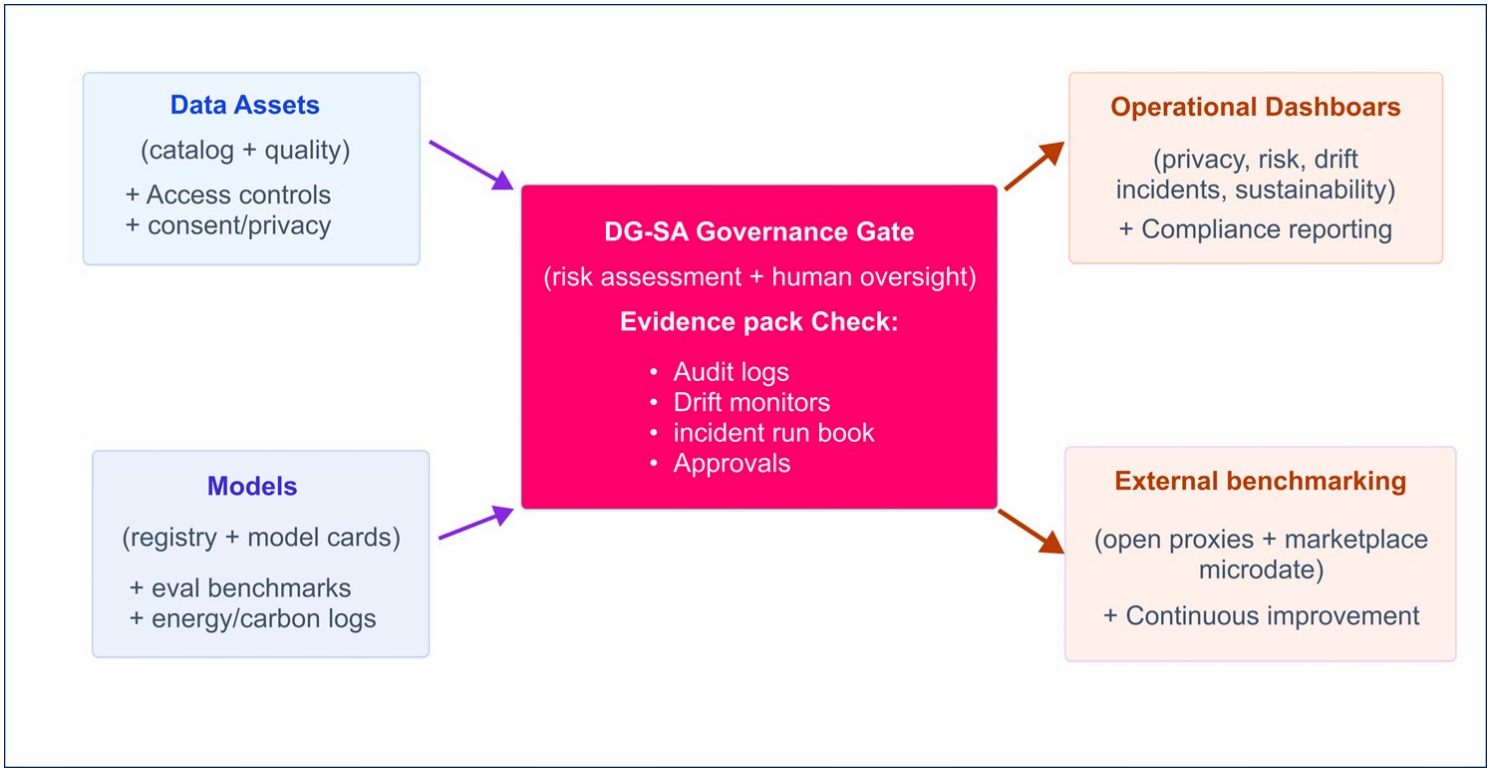

Figure 7

Mini-case instantiation: enterprise GenAI assistant governed via a DG-SA evidence gate and monitored using internal telemetry plus authenticated external indicators.

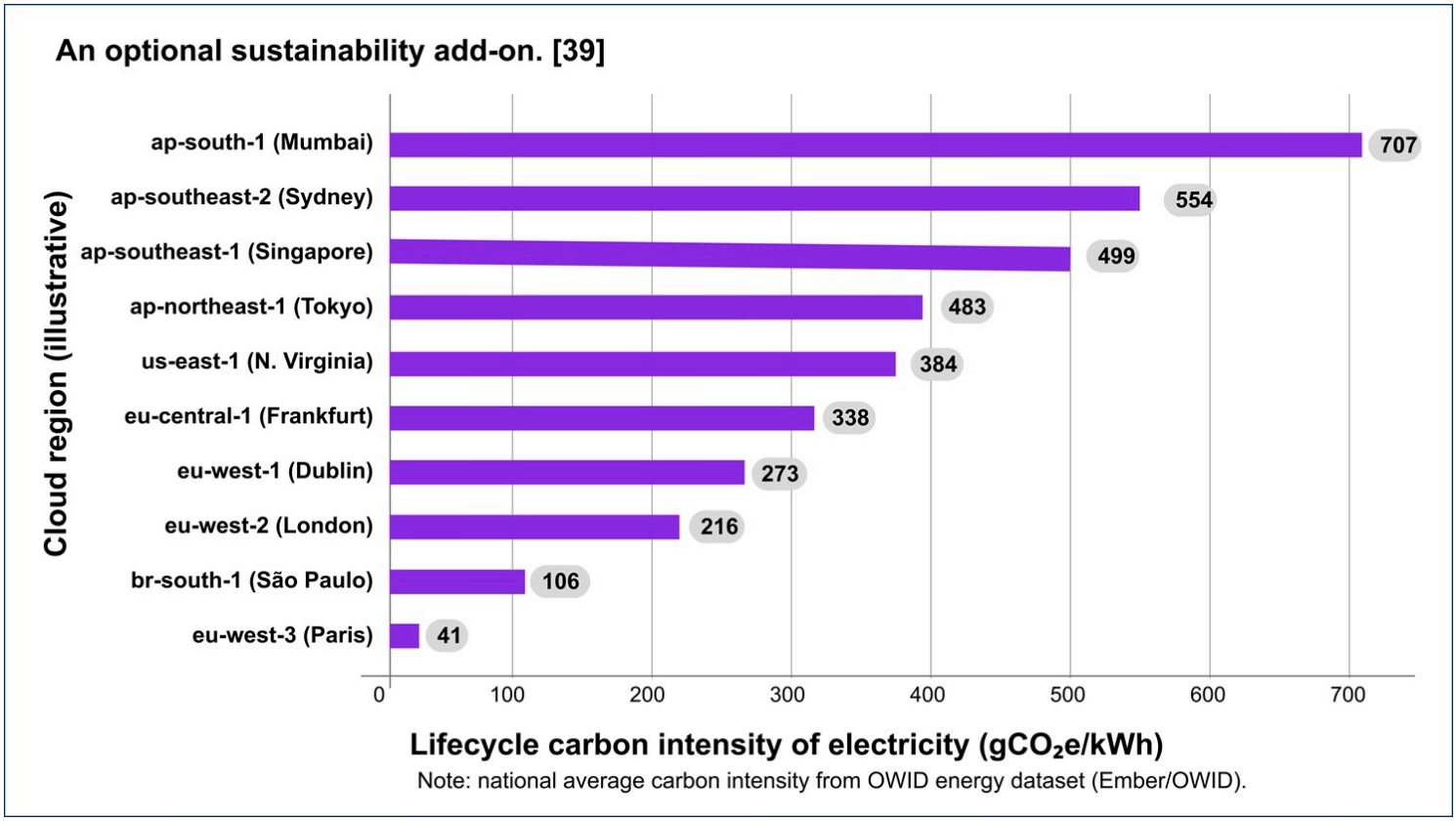

Figure 8

Carbon intensity of electricity across illustrative cloud regions (2024), using national-average OWID energy microdata (gCO_2e/kWh). [39].

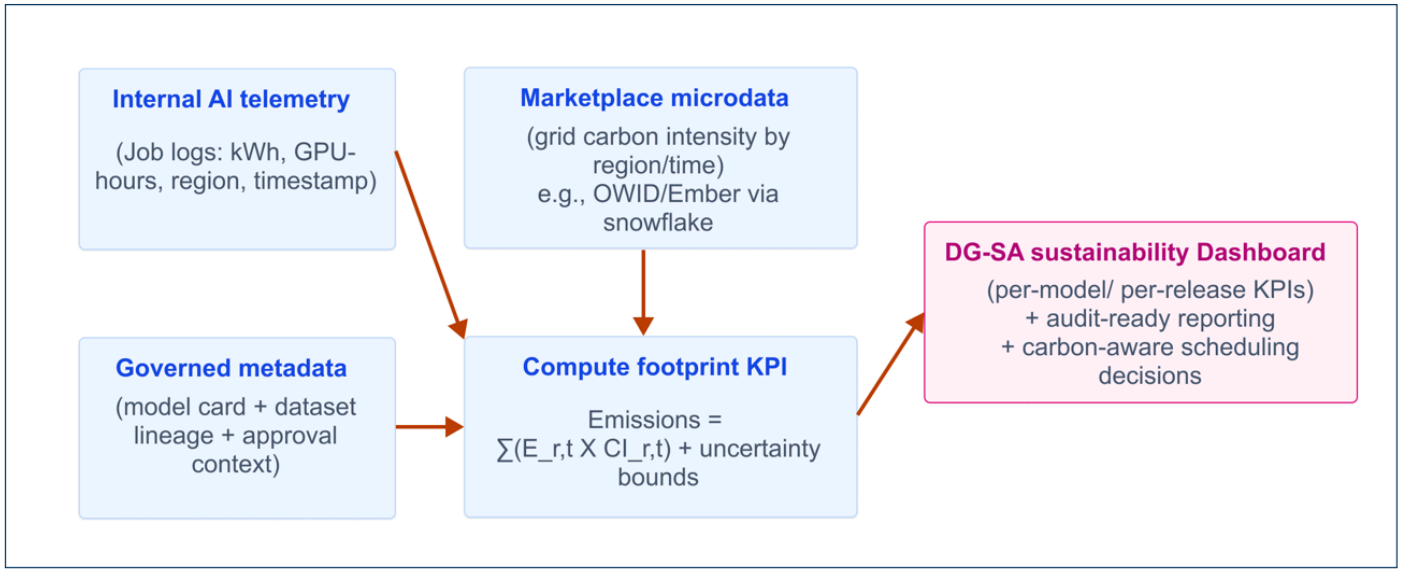

Figure 9

Marketplace microdata join pattern: combining governed AI telemetry and lineage with external carbon-intensity microdata yields an audit-ready sustainability KPI.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Graphicalabstract.jpg
- APPENDIX.docx